1 Bilinear Maps



Definition 1.1

Let V be a vector space over F. A *bilinear form* on V is a function $B : V \times V \to F$ which is linear in each variable separately when the other variable is held constant. That is, for all $v_1, v_2, v \in V$ and $\alpha \in F$, we have:

$$B(\alpha v_1 + v_2, v) = \alpha B(v_1, v) + B(v_2, v)$$

and

$$B(v, \alpha v_1 + v_2) = \alpha B(v, v_1) + B(v, v_2).$$



Exercise 1.2

Show that $B : F^2 \times F^2 \to F$ defined by

$$B((u, v), (s, t)) = ut - vs$$

is a bilinear form on F^2.



Solution

We have:

$$\begin{aligned} B((u, v), \alpha(s_1, t_1) + (s_2, t_2)) &= B((u, v)(\alpha s_1 + s_2, \alpha t_1 + t_2)) \\ &= u(\alpha s_1 + s_2) - v(\alpha t_1 + t_2) \\ &= \alpha(us_1 - t_1 v) + us_2 - tv_2 \\ &- \alpha B((u, v), (s_1, t_1)) + B((u, v), (s_2, t_2)), \end{aligned}$$

and the other direction follows in the same manner.



Exercise 1.3

Let $V = M^{n\times 1}(F)$. Show that if A is an $n \times n$ matrix, then

$$B_A(x, y) = x^T A y$$

is a bilinear form. Note that the bilinear form of the previous exercise is of this form with $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$.



Solution

We have:

$$\begin{aligned} B_A(\alpha x_1 + x_2, y) &= (\alpha x_1 + x_2)^T A y \\ &= (\alpha x_1^T + x_2^T) A y \\ &= \alpha(x_1^T A y) + x_2^T y \\ &= \alpha B_A(x_1, y) + B_A(x_2, y), \end{aligned}$$

and the other direction is obvious since multiplication by A and x^T are linear maps.

Exercise 1.4

Show that $B : \mathcal{P}_n(\mathbf{R}) \times \mathcal{P}_n(\mathbf{R}) \to \mathbf{R})$ defined by $B(p, q) = p(2)q'(3)$ is a bilinear form on $:P_n(\mathbf{R})$.

Solution

We have $B(\alpha p_1 + p_2, q) = (\alpha p_1 + p_2)(2)q'(3)$ at which point this follows from linearity of the evaluation map (and similarly for q.

Exercise 1.5

Let V be any vector space over F and fix $f, g \in V^*$. Show that $B : V \times V \to F$ defined by $B(u, v) = f(u)g(v)$ is a bilinear form. More generally, if $f_1, ..., f_n, g_1, ..., g_n \in V^*$, then $B(u, v) = \sum_{i=1}^{n} f_i(u)g_i(v)$ is a bilinear form on V.

Exercise 1.6

We have:

$$B(au_1 + u_2, v) = \sum_{i=1}^{n} f_i(au_1 + u_2)g_i(v) = \sum_{i=1}^{n} af_i(u_1)g_i(v) + f_i(u_2)v_i = a\sum_{i=1}^{n} f_i(u_1)g_i) + \sum_{i=1}^{n} f_i(u_2)g(v) = aB(u_1, v) + I$$

and the second component is identical.

Remark 1.7

Note that $V \times V$ is a vector space with respect to the operations

$$(u_1, v_1) + (u_2, v_2) := (u_1 + u_2, v_1 + v_2)$$
$$\alpha(u, v) := (\alpha u, \alpha v).$$

If $B : V \times V \to F$ is a bilinear map on V, we can then ask under what condition is B also a linear functional on $V \times V$? For such a B, we would have $B(\alpha u, \alpha v) = \alpha B(u, v)$ since B is linear as well as $B(\alpha u, \alpha v) = \alpha^2 B(u, v)$ since B is bilinear, which implies $(\alpha^2 - \alpha)B(u, v) = 0$ for all $u, v \in V$ and all $\alpha \in F$. For this to hold for $\alpha \neq 0, 1$, we must therefore have $B(u, v) = 0$ for all $u, v \in V$, hence B a bilinear map on V is also a linear functional on $V \times V$ if and only if B is the zero map (which is easily verified to be a bilinear form on V).

Exercise 1.8

Let $V^{(2)}$ denote the set of all bilinear forms on V. Show that this is a subspace of $F^{V \times V}$, the vector space of all maps $V \times V \to F$.

Proof:

The zero map is a bilinear form, so it suffices to show if B_1, B_2 are bilinear forms, so is $\alpha B_1 + B_2$.
We have:

$$(\alpha B_1 + B_2)(au_1 + u_2, v) = \alpha B_1(au_1 + u_2, v) + B_2(au_1 + u_2, v) = \alpha a B_1(u_1, v) + \alpha B_1(u_2, v) + aB_2(u_1, v) + B_2(u_2, v) =$$

and the other direction again follows similarly.



Suppose now that V is a finite-dimensional vector space with $\dim V = n$. We will now prove that $V^{(2)}$ is also finite-dimensional with $\dim V^{(2)} = n^2$.



Definition 1.9 (Matrix of a bilinear form)

Let B be a bilinear form on V and let $(e_1, ..., e_n)$ be an ordered basis of V. Define the *matrix of* B with respect to this ordered basis to be the matrix $[B]$ whose ij-entry is given by

$$[B]_{i,j} = B(v_i, v_j).$$

Note - a better way of thinking about this is considering a "dual basis" $e^*_{i,j}$ given by $e^*_{i,j}(e_p, e_q) = \delta_{ip}\delta_{jq}$. This is really the same thing, but it's fine.



Exercise 1.10

Find the matrix of the bilinear form $B : \mathcal{P}_2(\mathbf{R}) \times \mathcal{P}_2(\mathbf{R}) \to \mathbf{R}$ of Exercise 1.4, defined by $B(p, q) = p(2)q'(3)$ with respect to the standard ordered basis of $\mathcal{P}_2(\mathbf{R})$.



Solution

We have:

$$\begin{bmatrix} 0 & 1 & 6 \\ 0 & 2 & 12 \\ 0 & 4 & 24 \end{bmatrix}$$



Exercise 1.11

Prove that the map $T : V^{(2)} \to M^{n\times n}(F)$ defined by $T(B) = [B]$ is an isomorphism. This proves that $V^{(2)}$ is finite-dimensional with $\dim V^{(2)} = n^2 = (\dim V)^2$.



Proof:

Using the dual basis characterization (naturally isomorphic to $M^{n\times n}$ by dimension), it is clear the dual basis elements are bilinear forms themselves and independent (using essentially the same argument as for the regular dual space), so all that suffices to show is surjectivity. But, every bilinear map is defined by its value on the dual bases, and so this is clearly surjective and we are done. Really, this is the same argument as for the dual space.

